

강의계획서(2026학년도 2학기)

교과목명	융합정보전송	분반	1	담당교수명	에런스노버거
		학점	3	연락처	
교과목 코드	0000130650	요일/시간	수 1-A, 수 1-B, 수 2-A, 수 2-B, 수 3-A, 수 3-B	E-mail	aaronkr.trainer@gmail.com
교과목 구분	전공선택			연구실	
학과/학년	국제이공학부(엔지니어링사이언스) 4	강의실	전주:정보화 교육동 203	상담가능시간	

1. 강의 기본정보

수업목표	Information theory and inference, often taught separately, are here united in one entertaining textbook. This textbook introduces theory in tandem with applications.							
직전 강의평가 및 CQI 반영사항	This is a new class.							
6대 핵심역량과의 관계								
구분	소통역량	창의역량	인성역량	실무역량	모험역량	문화역량	합계	대표역량
비율(%)	10%	20%	0%	50%	20%	0%	100%	
교과목간의 연계성								
교재언어	영어	강의언어	영어	필요기자재				
교재 목록								
교재구분	ISBN번호	도서명		저자명	출판사명	출판년도		
주교재		Information Theory, Inference and Learning Algorithms		MacKay, David J. C.	Cambridge University Press	2003년 10월 06일		
권장선수과목				권장후속과목				
수업방식 (복수가능)	강의	실험/실습	설계	발표/토론	과제	Quiz	프로젝트	
	☒	☒	☐	☐	☐	☐	☐	
	발표	현장학습	기타	AIX-PBL	PBL	플립러닝	LMS활용	
	☐	☐	☐	☐	☐	☐	☐	
수업운영방향								
AI 활용방법	생성형 AI 사용 금지		생성형 AI 사용 부분적 허용		생성형 AI 사용 전면적 허용			
	☐		☒		☐			
AI 활용 세부내용	My full AI policy is on my website: https://courses.aaron.kr/policies/ai/							
평가계획 (100%)	중간	기말	출석	과제물	안전교육	발표/토론	수업태도	기타
	25%	25%	20%	30%	0%	0%	0%	0%
평가참고사항								
평가방법	절대평가	상대평가 비율	A(%)	A+B(%)	C이하(%)	총비율		
		절대평가 기준	절대평가					
* 장애학생 교수학습지원 사항								

참고사항	☒	강의 파일, 자료 등 제공	☐	좌석배치(지정좌석 조정)	☐	기타	
	☒	과제 제출기간 연장	☐	대안적 과제의 제시			
	☒	시험기간 연장	☐	평가방법 조정(예 : 대독, 구두 응답, 도우미 대필 답안 작성 등)			
	☐	별도의 시험 장소 제공	☐	기타			
	☐	그 외(해당하는 내용 자유로이 기술)					
※ 위 지원사항 등을 포함한 강의, 과제, 시험 등 학습과정에서 장애로 인하여 추가 지원이 필요한 경우 개강전 담당강사 및 장애학생지원센터를 통해 문의 바랍니다.							
주별 내용							
주별	수업목표	수업내용	수업방식	자료, 과제 및 기타 참고사항	수업방식별시간		
					온라인	오프라인	
1주차	Introduction	Class / grading overview, homework types & submission methods	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
2주차	0. Basic Git usage 1. Introduction to Information Theory 2. Probability, Entropy, and Inference 3. More about Inference	0. Basic Git usage 1. Introduction to Information Theory 2. Probability, Entropy, and Inference 3. More about Inference	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
3주차	I: Data Compression 4. The Source Coding Theorem 5. Symbol Codes 6. Stream Codes 7. Codes for Integers	I: Data Compression 4. The Source Coding Theorem 5. Symbol Codes 6. Stream Codes 7. Codes for Integers	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
4주차	II: Noisy-Channel Coding 8. Dependent Random Variables 9. Communication over a Noisy Channel 10. The Noisy-Channel Coding Theorem 11. Error-Correcting Codes and Real Channels	II: Noisy-Channel Coding 8. Dependent Random Variables 9. Communication over a Noisy Channel 10. The Noisy-Channel Coding Theorem 11. Error-Correcting Codes and Real Channels	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
5주차	III: Further Topics in Information Theory 12. Hash Codes: Codes for Efficient Information Retrieval 13. Binary Codes 14. Very Good Linear Codes Exist 15. Further Exercises on Information Theory	III: Further Topics in Information Theory 12. Hash Codes: Codes for Efficient Information Retrieval 13. Binary Codes 14. Very Good Linear Codes Exist 15. Further Exercises on Information Theory	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
6주차	III: Further Topics in Information Theory 16. Message Passing 17. Communication over Constrained Noiseless Channels 18. Crosswords and Codebreaking 19. Why? Information Acquisition and Evolution	III: Further Topics in Information Theory 16. Message Passing 17. Communication over Constrained Noiseless Channels 18. Crosswords and Codebreaking 19. Why? Information Acquisition and Evolution	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
7주차	IV: Probabilities and Inference 20. A Example Inference Task: Clustering 21. Exact Inference by Complete Enumeration 22. Maximum Likelihood and Clustering 23. Useful Probability Distributions	IV: Probabilities and Inference 20. A Example Inference Task: Clustering 21. Exact Inference by Complete Enumeration 22. Maximum Likelihood and Clustering 23. Useful Probability Distributions	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
8주차	Midterm Test / Project / Report	Midterm Test / Project / Report	Midterm Test / Project / Report	Midterm Test / Project / Report			
9주차	IV: Probabilities and Inference 24. Exact Marginalization 25. Exact Marginalization in Trellises 26. Exact Marginalization in Graphs 27. Laplace's Method	IV: Probabilities and Inference 24. Exact Marginalization 25. Exact Marginalization in Trellises 26. Exact Marginalization in Graphs 27. Laplace's Method	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
10주차	IV: Probabilities and Inference 28. Model Comparison and Occam's Razor 29. Monte Carlo Methods 30. Efficient Monte Carlo Methods 31. Ising Models 32. Exact Monte Carlo Sampling	IV: Probabilities and Inference 28. Model Comparison and Occam's Razor 29. Monte Carlo Methods 30. Efficient Monte Carlo Methods 31. Ising Models 32. Exact Monte Carlo Sampling	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
11주차	IV: Probabilities and Inference 33. Variational Methods 34. Independent Component Analysis and Latent Variable Modelling 35. Random Inference Topics 36. Decision Theory 37. Bayesian Inference and Sampling Theory	IV: Probabilities and Inference 33. Variational Methods 34. Independent Component Analysis and Latent Variable Modelling 35. Random Inference Topics 36. Decision Theory 37. Bayesian Inference and Sampling Theory	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
12주차	V: Neural Networks 38. Introduction to Neural Networks 39. The Single Neuron as a Classifier 40. Capacity of a Single Neuron 41. Learning as Inference	V: Neural Networks 38. Introduction to Neural Networks 39. The Single Neuron as a Classifier 40. Capacity of a Single Neuron 41. Learning as Inference	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
13주차	V: Neural Networks 42. Hopfield Networks 43. Boltzmann Machines 44. Supervised Learning in Multilayer Networks 45. Gaussian Processes 46. Deconvolution	V: Neural Networks 42. Hopfield Networks 43. Boltzmann Machines 44. Supervised Learning in Multilayer Networks 45. Gaussian Processes 46. Deconvolution	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
14주차	VI: Sparse Graph Codes 47. Low-Density Parity-Check Codes 48. Convolutional Codes and Turbo Codes 49. Repeat-Accumulate Codes 50. Digital Fountain Codes	VI: Sparse Graph Codes 47. Low-Density Parity-Check Codes 48. Convolutional Codes and Turbo Codes 49. Repeat-Accumulate Codes 50. Digital Fountain Codes	PPT slides & Python	Colab / GitHub notebooks, submission via GitHub			
15주차	Final Test / Project / Report	Final Test / Project / Report	Final Test / Project / Report	Final Test / Project / Report			